INES-RUHENGERI · FACULTY OF SCIENCES AND INFORMATION TECHNOLOGY

# MyTurn

## Design and Implementation of an IoT-Based Waiting Line Management and Service-Desk Status System

**Your Turn, Made Simple.**

TICKET PAGE | WALL DISPLAY | DESK CONSOLE | ESP32 BOARD


| | |
|---|---|
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Internet of Things (IoT)
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# A numbered line is still a line.

A visitor walks in, takes a number, and is told to wait. The number says where they are in the line, and nothing else: not how long the wait is, not whether the desk is open, not whether the desk is still calling numbers at all.

1

### The wait is invisible

A ticket cannot tell a visitor whether A014 means five minutes or forty. So people crowd the counter, drift away, or come back after their number was called. **The line is anxious, and the counter is empty.**

2

### One absence stalls the line

The desk calls a number that nobody answers. With no rule to recover it, the service point stands idle while everyone behind that number waits. **One visitor who is not there holds up the queue.**

3

### Nothing is measured

A paper list cannot say how long a service took, how many visitors never answered, or who changed the desk. **Without a record, next week repeats this week.**

**The problem, stated plainly.** A visitor cannot see the state of the line they joined, and the desk has no rule that recovers when the visitor it called does not come. MyTurn closes both gaps: a live status for the visitor, and a queue rule that returns an unanswered call to its own line.

# A QR code, a number, and a desk that recovers by itself.

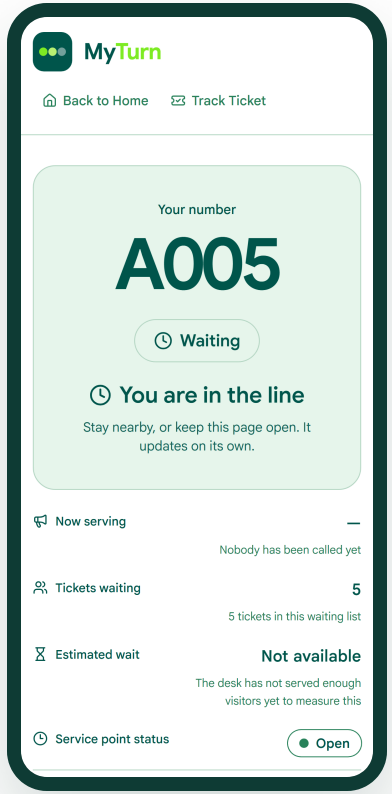
1 **Scan at the door**
The QR code opens the ticket page in the visitor's own phone. No app, no account.

2 **Choose the service**
The server issues the next number of that waiting line and a private tracking link.

3 **Leave a way to be called**
One email address or phone number, so the desk can reach the visitor when their turn comes.

4 **Follow your turn**
The page shows the called number, the wait and how many are ahead, and it updates on its own.

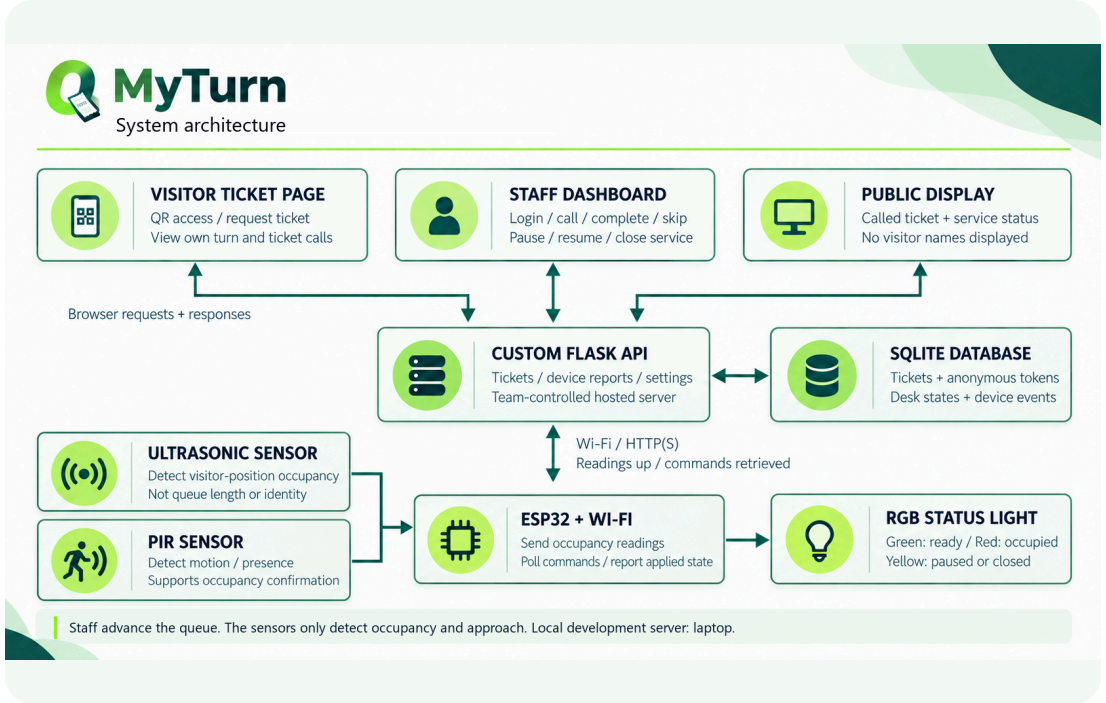
**No name · no account · one contact, used only for the call**




The visitor page, built and running

# Four parts, one waiting line.

- **The board measures**
  An ESP32 reads distance and approach, then applies the desk state to four LEDs.
- **The API decides**
  Flask owns the waiting line rules, a counter per service, the desks, the accounts and the event log.
- **SQLite remembers**
  Services, waiting lines, desks, board readings and accounts in one file, with every change written down.
- **The pages show**
  Next.js: visitor, wall display, staff console and administration.
- **One origin**
  The browser reaches the API through the same origin; the board proves itself with a device key.




ESP32 → Flask API → SQLite → Next.js

# What sits at the service point.

| | |
|---|---|
| **ESP32 DevKitC** | the controller |
| **HC-SR04 ultrasonic** | TRIG 5 · ECHO 21 |
| **HC-SR501 motion** | GPIO 19 |
| **Four LEDs** | 25 · 26 · 27 · 33 |
| **Buzzer** | GPIO 32 |

**RWF 46,200** estimated prototype budget

The enclosure, the sensor mount and the power supply are part of the same budget.




Wiring at a glance · the full schematic is in the report

# The sensors measure a place, not a person.

### MEASURED

- ✓ Distance from the HC-SR04, turned into **occupied** or **free**.
- ✓ An approach from the PIR, held for a short window.
- ✓ The LED state the board actually applied, sent back with every report.
- ✓ A median of five distance readings, trusted only after three, so one bad echo does not change the state.
- ✓ Each reading is stored with its time, distance, occupancy, approach and applied LED state.

### NEVER

- × No counting people in a line.
- × No identity, no face, no name.
- × No camera, no microphone, no tracking.
- × No guess about who deserves priority.
- × No record of a person, a visit or a conversation.

A waiting line system does not need to know who you are. The HC-SR04 alone decides occupancy; the PIR only reports that somebody is walking up.

**A bag is not a person.** An object that has not moved for five minutes, with no approach seen in that window, stops blocking the call. The hold rule is not relaxed: only the call guard is.

# One line per service, one call per desk.

1. **The number comes from the server**
   Each service has its own counter row, updated in the same transaction that inserts the ticket, so two requests cannot share a number.

2. **The link is opaque**
   The tracking URL carries a random token, never the ticket number or a sequential id.

3. **One call per desk**
   A desk serves one service and holds one call; two desks may work the same service at once. A per-desk unique index and a transaction enforce it.

4. **Nobody blocks the desk**
   After the hold, an unanswered number goes to the back of its own service with a fresh number.

5. **Everything is written down**
   Every change is an event with a time and an actor: a person, or the system.

## A TICKET'S LIFE

**waiting** called → **called** completed → **completed**

**called** skipped → **skipped**

**called** cancelled → **cancelled**

**called** hold ends → **waiting** fresh number

recall restarts the hold; undo takes back the last decision within 10 seconds. The default hold is 60 seconds, and the next slide shows the one case where the board can shorten it.

# The board can end an unanswered call early. Only with evidence.

**Default hold** 60 seconds




Without a fresh reading, the full hold always applies.

**With evidence** 25 seconds




The unused part of the hold is given back to the waiting line.

grace · service point empty · full hold · not used

✓ **Fresh board** ✓ **No occupancy** ✓ **No approach** ✓ **Past the grace** **Otherwise: full hold**

The server decides. The board only reports. An offline board, a missing reading, an occupied service point or an approach keeps the full hold, so absence of data is never treated as absence of a person.

**Why a floor?** A visitor who is walking in needs time to reach the desk. Ten seconds of grace, then a floor of 25, stops the rule from firing before they arrive.

# A wait you can trust, or no number at all.

**3**

served tickets before the page offers an estimate.
Below that it says “Not available”.

MYTURN_ETA_MIN_SAMPLES, not a guess

**estimate** = measured average × tickets ahead

### 1 Measured, not modelled

Each sample is the time from **called_at** to **closed_at** of a ticket that was actually served.

### 2 The sample count is visible

The visitor sees the average and how many tickets it came from, so a thin average cannot look certain.

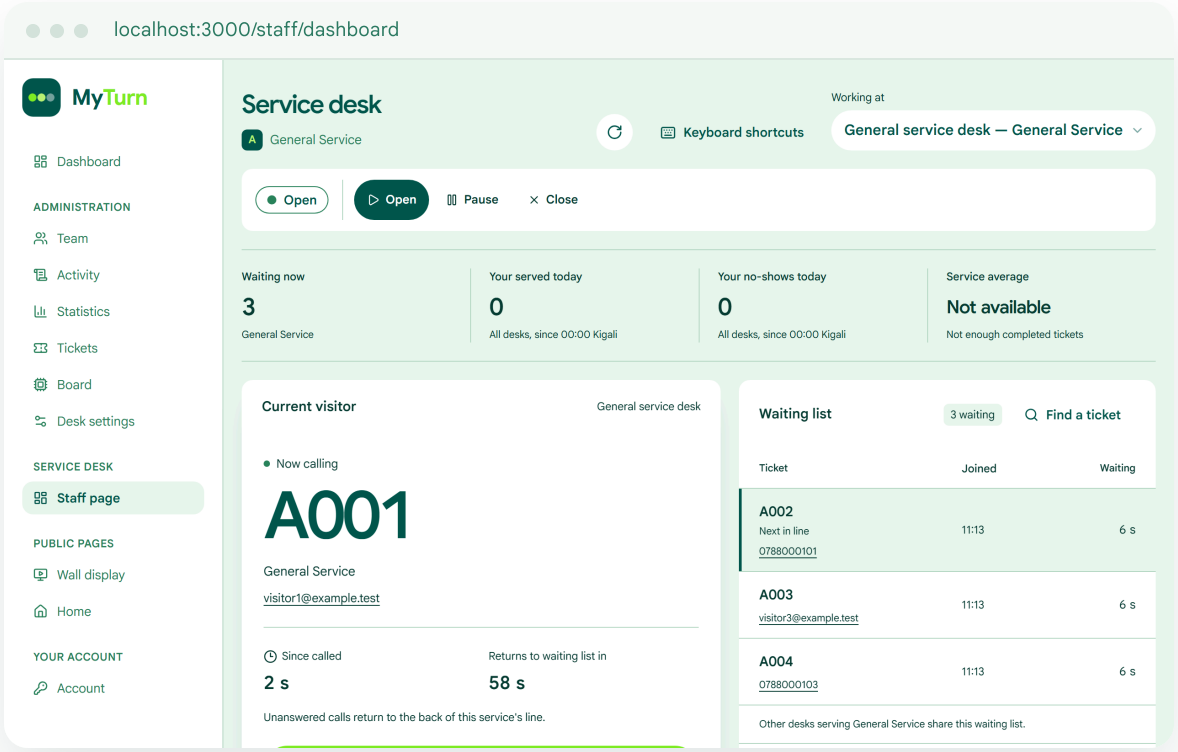
### 3 Small demo times are shown

A desk that serves in seconds reports seconds. The figure is not hidden to look better.

### 4 A target is compared, not promised

If the service sets a target, the statistics page reports how often it was actually met.

# Two views of the same waiting line.




**Staff:** call, served, no-show, cancel, walk-in, recall and undo.

**Live:** hold countdown, the board's readings, keyboard shortcuts, today's tallies.

## Administration

**A — Team & invitations**
Invite, promote, switch off; the last administrator cannot be removed.

**B — Activity**
Ticket, desk and account history in one list, with the reason for each requeue.

**C — Statistics**
Where every ticket went, the busiest hours, how long visitors waited, and the CSV of the period.

**D — Tickets, board and settings**
Search every ticket, register the sensor board watching a counter, and edit each service and service point.

# Rights are granted, checked and recorded.

VISITOR

### No account

An opaque ticket token in the link, and one contact the visitor types so the desk can call them. No name, no profile.

STAFF

### Invitation

Single-use, 72 hours. Sign in as a person, or with the shared classroom password.

ADMINISTRATOR

### Promoted

Created only by the seed script or by another administrator. Never by public registration.

DEVICE

### Device key

The ESP32 sends a key with every reading. It can report and ask for a setting; it cannot change the waiting line.

**Server-side roles.** Every admin route is checked twice: the page, and the API.

**Invitations cannot create admins.** Asking for one is a 400.

**No address leaks.** A password reset answers the same whether or not the address exists, and a visitor's contact never reaches the wall display.

**CSRF on writes.** A double-submit token, kept in memory, not in localStorage.

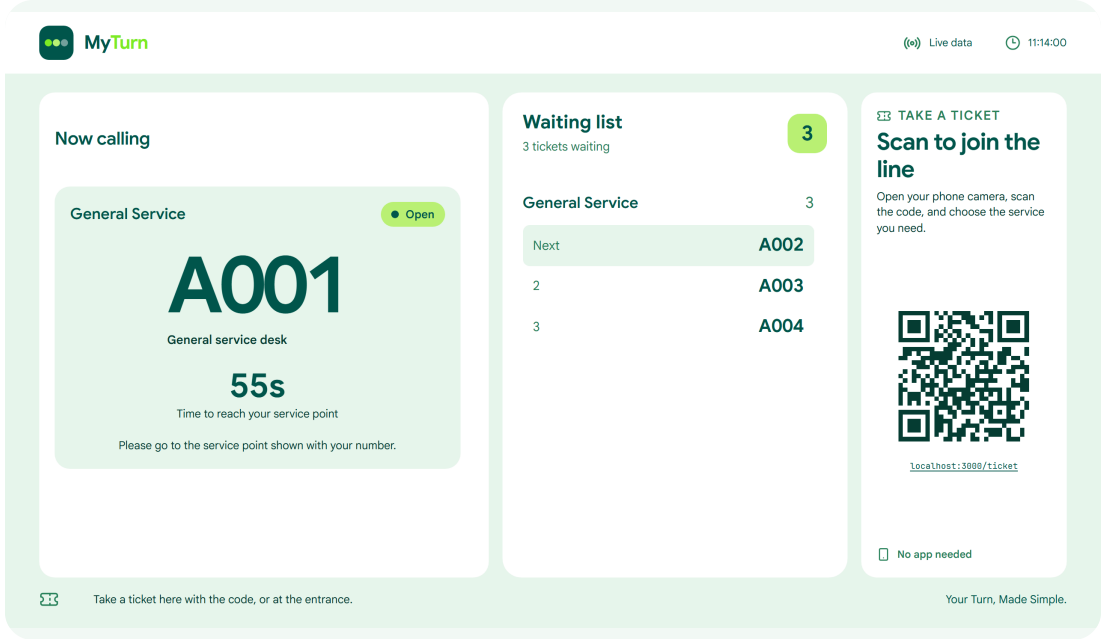
**Guarded last admin.** You cannot demote the final administrator or switch off your own account.

**Recorded.** Account changes land in the activity history with the actor and the time.

> A visitor never gets an account: a ticket carries a number, a token, and one way to reach them.

# What we will show on the day.

1. **Take a ticket on a phone**
   Pick a service and leave a phone number or an email; the number comes from that line's counter and the tracker opens.
2. **Watch the call arrive**
   Call next at the desk; the wall screen and the phone show the same number, without a refresh.
3. **Serve or record a no-show**
   Complete, skip or cancel. Undo is available for ten seconds.
4. **Let a call expire**
   The ticket goes to the back of its service with a fresh number, and the reason is logged.
5. **Show the board**
   Distance and approach change the LED; the dashboard shows what the board applied.
6. **Close the desk**
   The ticket page says so before a visitor joins, and the LED follows.




The waiting-area display

# Tested, not assumed.

**234**
API checks: accounts, CSRF, hold, requeue, estimate gate, services and desks

**71**
browser checks on the visitor, auth and desk pages

**45**
console checks: activity, statistics, settings, roles

**8**
sidebar checks, including the shared-session rules

Every check was run against the running prototype, not read off the source.

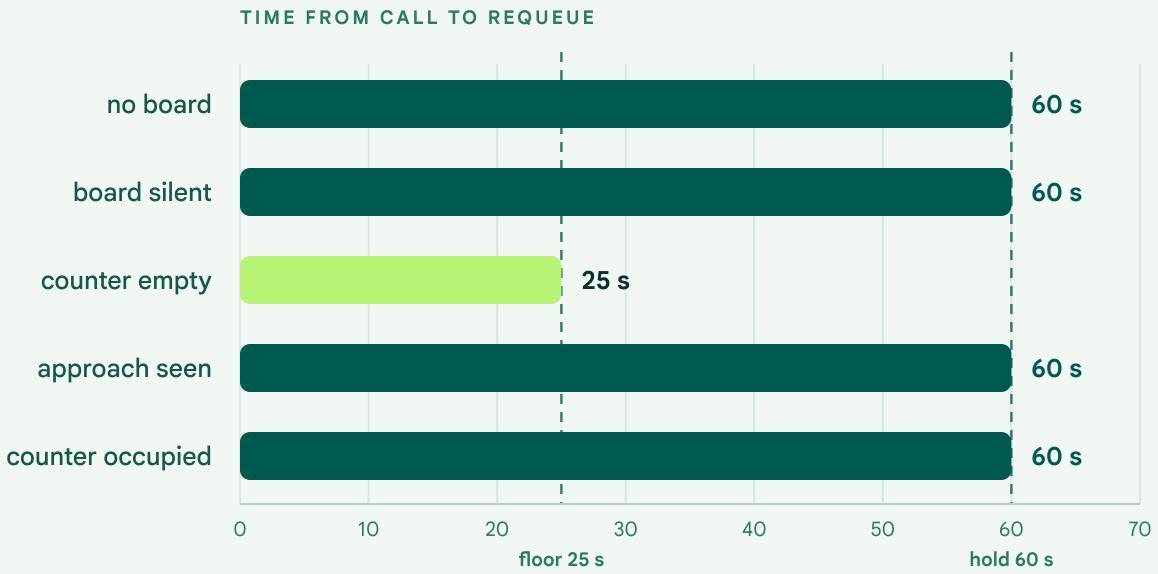
✓ **The build is clean**
TypeScript, ESLint and the production build all pass; all sixteen pages are generated.

✓ **The rules are exercised**
Auth and CSRF, the occupancy requeue with synthetic readings, the estimate gate, undo, recall, CSV, and the board's WebSocket reporting.

! **Still measured by hand**
The physical sensor trials and the photograph of the desk rig are the remaining figures; the report leaves them blank rather than inventing them.

358 automated checks. A feature is only finished when a check proves it.

# The board earns thirty-five seconds.




Measured from ticket_events: requeue time minus call time, five calls at five desks, same moment

**60 s**

### Four scenarios, no shortening

No board at all, a board that went quiet before the call, a visitor approaching, a visitor standing at the counter.

**25 s**

### One scenario, shortened

A report after the call, the counter empty, and nothing approaching for the whole window.

**35 s**

### What the waiting line gains

The next number is called 35 seconds earlier, and no desk ever has to trust a silent sensor.

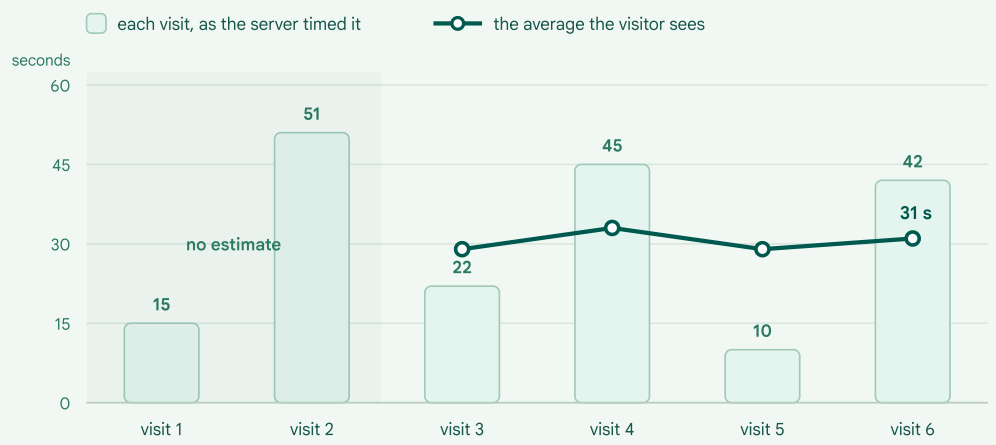
**1 desk**

### Evidence is per counter

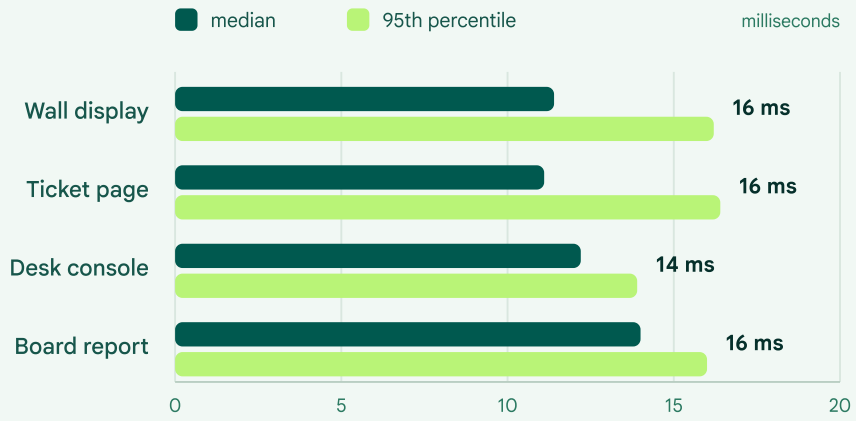
A board only shortens the call at its own desk; a reading from the counter next door never counts.

One case in five: only a fresh reading of an empty counter shortens the hold. Every other case keeps the full minute.

# No guess, no wait.




Six visits of 15 to 51 seconds. Nothing is offered until three are served; from the third, the figure tracks the mean to within half a second.




Forty requests per endpoint against the running API. Every read a page or a board makes answers in 11 to 14 ms at the median, under 17 ms at the 95th percentile.

**One controlled run.** Eleven tickets in a disposable office, the shipped timings, nothing live touched; latency measured on this machine serving itself, not on a school network.

# What it does not do yet.

### No daily reset

The per-service counter keeps climbing across days. A midnight reset is specified but not built.

NOT BUILT

### No separate SERVING state

A called ticket is treated as being served; the dwell time is the service time. This is documented, not invented.

DELIBERATE

### Sensors measure only

The PIR can see a passer-by, and the ultrasonic sensor needs a clean mount. Field accuracy is still to be recorded.

TO MEASURE

### No buffering on the board

An offline board keeps the full 60-second hold, by design. Readings taken while the Wi-Fi is down are discarded, not replayed.

NOT BUILT

### Notifications not yet delivered

Email and SMS calls are wired to SMTP and TextBee and tested with mocked transports, but no provider is configured, so nothing has left this machine.

UNVERIFIED

### Single process

SSE subscribers and rate-limit counters live in memory, so the API runs one worker. Scaling needs shared state.

DEPLOYMENT

Also not built: spoken announcements. Invitation and reset links still appear in the app rather than an inbox.

# Delivered, and what comes next.

## DELIVERED

- ✓ A working prototype: 16 pages, 12 tables, real accounts and roles.
- ✓ The ESP32 sketch: distance, approach, four LEDs, the buzzer and the desk setting.
- ✓ Four suites, 358 automated checks, all passing.
- ✓ A proposal, a 55-page report and the design documents.

## NEXT

1. Run and record the physical sensor trials; add the desk photograph.
2. Fill the measured figures into the report, without estimating them.
3. Buffer readings on the board when the Wi-Fi drops.
4. Add daily number resets for every service.
5. Speak the number aloud for visitors who cannot see the screen.

A working prototype, a tested waiting line, and a clear list of what still needs measuring.

**Thank you. Questions?**

Your Turn, Made Simple.